

BUSINESS REQUIREMENTS DOCUMENT

HR Employee Attrition Analytics Dashboard

Prepared using Power BI, DAX, and Power Query

Field	Details
Document Owner	Priyanshu Kumar
Project Name	HR Employee Attrition Analytics Dashboard
Data Source	IBM HR Analytics Employee Attrition Dataset (Fictional dataset by IBM Data Scientists)
Tool Used	Microsoft Power BI Desktop
Document Version	1.0
Status	Final

1. Purpose of the Document

This Business Requirements Document (BRD) defines the objectives, scope, stakeholders, and functional requirements for the HR Employee Attrition Analytics Dashboard. The dashboard is designed to help HR and business stakeholders understand workforce composition, identify key drivers of employee attrition, and support data-driven retention decisions using interactive Power BI reporting.

2. Project Background

Employee attrition directly impacts organizational cost, productivity, and morale. Without a centralized view of attrition patterns, HR teams often rely on scattered reports and manual analysis to understand why employees leave. This project consolidates employee data — including demographics, compensation, satisfaction, and work-life balance factors — into a single interactive dashboard, enabling faster and more informed HR decision-making.

The dataset used is IBM's publicly available HR Analytics Employee Attrition dataset, a fictional dataset created by IBM data scientists for analytics practice and research purposes.

3. Business Objectives

- Identify the key factors contributing to employee attrition across departments, job roles, and demographics.
- Provide HR stakeholders with a self-service dashboard to explore attrition trends without manual report requests.
- Enable comparison of compensation, satisfaction, and work-life balance metrics against attrition outcomes.
- Support proactive retention strategy planning through visual, insight-driven reporting.
- Reduce time spent on manual HR MIS reporting by centralizing key workforce KPIs in one place.

4. Project Scope

4.1 In Scope

- Analysis of employee demographic data (age, gender, marital status, education).
- Analysis of attrition patterns across department, job role, business travel, and overtime status.
- Compensation analysis including salary distribution by department, job role, gender, and attrition status.
- Employee satisfaction analysis covering job satisfaction, environment satisfaction, relationship satisfaction, and work-life balance.
- Interactive filtering via slicers (Age Group, Gender) and cross-page navigation.

4.2 Out of Scope

- Real-time or live data integration (the dataset is static and refreshed manually).
- Predictive modeling or machine learning-based attrition forecasting.
- Integration with live HRMS/payroll systems.

- Employee-level personally identifiable reporting (analysis is conducted at an aggregate/category level).

5. Stakeholders

Stakeholder	Role / Interest
HR Manager / HR Business Partner	Primary consumer; uses dashboard to identify attrition drivers and plan retention actions
Department Heads	Reviews attrition and compensation trends specific to their department
Senior Management / Leadership	Uses high-level KPIs for workforce planning and budget decisions
Dashboard Developer (Analyst)	Designs, builds, and maintains the Power BI report

6. Functional Requirements

The dashboard is organized into five pages, each addressing a specific business question:

Page	Purpose	Key Elements
Overview	High-level snapshot of workforce health	KPI cards (Total Employees, Active Employees, Employees Left, Average Age, Average Monthly Income, Average Years at Company); navigation buttons to detail pages
Workforce Demographics	Understand workforce composition	Employees by Gender, Overtime, Marital Status (donut charts); Employees by Education, Age Group (column charts)
Attrition Analysis	Identify where attrition is concentrated	Attrition vs Business Travel, Job Role, Department, Gender, Overtime, Work-Life Balance; Attrition Rate KPI card; Age Group and Gender slicers
Compensation Analysis	Evaluate pay equity and its link to attrition	Salary vs Department, Job Role, Attrition, Gender; Median Salary and Salary Range KPI cards; synced slicers
Employee Satisfaction Analysis	Understand the human factors behind attrition	Job, Environment, Relationship Satisfaction vs Attrition; Work-Life Balance vs Attrition (attrition rate %, sorted by ordinal scale)

6.1 Navigation Requirements

- Users must be able to navigate from the Overview page to any of the four analysis pages using on-screen action buttons.

6.2 Filtering Requirements

- Age Group and Gender slicers must be available and synced across the Attrition Analysis and Compensation Analysis pages, so filter selections persist as users move between them.

7. Data Requirements

Source: WA_Fn-UseC_-HR-Employee-Attrition dataset (IBM), containing employee-level records across demographic, job, compensation, and satisfaction attributes.

7.1 Key Derived Measures (DAX)

Measure	Definition
Total Employees	Total count of employee records
Employees Left	Count of employees where Attrition = "Yes"
Active Employees	Total Employees minus Employees Left
Attrition Rate (%)	Employees Left as a percentage of Total Employees, calculated per category (e.g., department, job role, satisfaction level)
Average Monthly Income	Average of MonthlyIncome across the filtered employee base
Average Years at Company	Average tenure of employees in years

7.2 Ordinal Field Handling

Fields with a natural ranked order — Education, Environment Satisfaction, Job Involvement, Job Satisfaction, Performance Rating, Relationship Satisfaction, and Work-Life Balance — are sorted using Power BI's "Sort by Column" feature against their original numeric codes (e.g., Work-Life Balance: 1-Bad, 2-Good, 3-Better, 4-Best), ensuring charts display categories in logical rather than alphabetical order.

8. Non-Functional Requirements

- Usability: Dashboard should be self-explanatory for a non-technical HR audience, with clear titles and consistent visual design.
- Performance: All visuals should render within 2-3 seconds on filter/slicer interaction.
- Consistency: All pages should follow a uniform color theme, card style, and layout grid.
- Accessibility: Font sizes and color contrast should be legible when presented on screen or exported to PDF.

9. Assumptions and Constraints

- The dataset is static (fictional, IBM-provided) and does not require scheduled refresh or live connectivity.
- All monetary values are assumed to be in the currency format used in the source dataset.
- Analysis is limited to the fields available in the dataset; no external data enrichment is performed.

10. Success Criteria

- Dashboard clearly identifies at least three measurable factors most strongly associated with attrition (e.g., low job satisfaction, frequent overtime, poor work-life balance).

- HR stakeholders can independently filter and explore data without needing a custom report for each query.
- Dashboard loads and performs smoothly across all five pages with no broken visuals or incorrect titles.

11. Appendix

11.1 Glossary

Term	Meaning
Attrition	An employee leaving the organization (voluntary or involuntary)
DAX	Data Analysis Expressions — the formula language used in Power BI for calculations
Slicer	An interactive filter control in Power BI
KPI	Key Performance Indicator